

Eda Gjergo, Ph.D.

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Publications

Web: <https://www.edagjergo.com>
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Research interests Galactic chemical evolution, initial mass function, dust modeling, cosmological simulations, interstellar and intracluster medium, supernova cosmology, dark energy models.

Current employment **Nanjing University** Nanjing, China
Assistant Research Fellow Sep 2025 – Present

Past employment **Nanjing University** Nanjing, China
Postdoctoral Scholar in Astronomy Sep 2022 – Aug 2025
Wuhan University Wuhan, China
Postdoctoral Scholar in Physics Oct 2019 – Aug 2022
INAF Trieste Observatory Trieste, Italy
Research Assistant Oct 2015 – May 2019
Argonne National Laboratory Lemont, IL, USA
Junior Team Member Sep 2011 – Aug 2015
Sperling & Kupfer Milan, Italy
Outreach author Jul 2005 – Jul 2007

Education **Università degli Studi di Trieste** Trieste, Italy
PhD in Physics Graduated in Spring 2019
Illinois Institute of Technology (IIT) Chicago, IL, USA
BS in Physics Graduated in Spring 2014
Illinois Institute of Technology (IIT) Chicago, IL, USA
BS in Applied Mathematics Graduated in Spring 2014
United World College (UWC) Duino, Italy
International Baccalaureate (IB diploma programme) Spring 2008

Teaching experience **Invited Lecturer (Charles University, Prague, Czech Republic)**
Selected topics in Astrophysics, 6 hours Winter 2025-2026
Invited Lecturer (Nanjing University, Nanjing, China)
Introduction to Galactic Chemical Evolution, 2 hours Spring 2023
Invited Lecturer (Wuhan University, Wuhan, China)
Classical Mechanics, 2 hours Fall 2020
Invited Lecturer (University of Trieste, Trieste, Italy)
General Relativity & Cosmology, 2 hours Spring 2017/2018
Invited Lecturer Yerkes Observatory Summer Program
(by the University of Chicago, Chicago, IL, USA)
Introduction to Cosmology, 1 hour Summer 2013

Honors and scholarships	Natural Science Foundation of China (NSFC) 56k USD	2026-2027
	National Institute of Astrophysics Fellowship (INAF)	2015-2018
	Chambliss Astronomy Achievement Student Award (AAS)	2013
	Merit-based International Scholarship (IIT)	2008-2014
	Winner of the Frascati Scienza National Award	2010
	Italian Ministry of Foreign Affairs Full Scholarship (UWC) (3.5% acceptance rate)	2006-2008
Mentoring	5. Soonchul CHOI, Postdoc, IBS, Daejeon, South Korea	2024-
	4. Hao-Yang LI, Nanjing University Undergraduate	2024-
	3. Zi-Hao LIN, Nanjing University undergraduate -> Nanjing University graduate student. Earned a top-tier grade and a 10,000 RMB research prize for his undergraduate research with me	2022-2023
	2. Run-Dong YU, Nanjing University undergraduate -> Johns Hopkins University graduate student	2022-2023
	1. Jin-Ning LIANG, Wuhan University undergraduate -> Durham University graduate student	2021-2023
Skills	Public Software I Developed and Released: GalCEM, pyIGIMF	
	Programming: Python, C, Mathematica, MATLAB, C++, IDL, Fortran90/77.	
	Specialized software: GADGET-2/3, CosmoMC, SNANA, SNCosmo	
	Editor/Misc.: Emacs, Vi, SVN, TotalView, Anaconda	
	Markup: HTML/CSS, $\LaTeX$	
Service and outreach	Languages: (native) Italian, English, (beginner) Chinese, French, German, Latin	
Service and outreach	Non-fiction bestseller <i>Così Parlano Le Stelle</i>	Apr '07
	YouTube Astrophysics Outreach: TheCosmicWeb	Dec '12 – Feb '14

Publications, Codes, and Talks

Monographs and Reviews

2. The initial mass function of stars
Kroupa P., Gjergo E., Jerabkova T., Yan Z. 2026
Encyclopedia of Astrophysics, Volume 2, ISBN 978-0-443-21439-4, Elsevier, 2026, pp. 173-210

1. The many tensions with dark-matter based models and implications on the nature of the Universe
Kroupa P., Gjergo E., et al.; 2023
Proceedings of Corfu2022: Workshop on Tensions in Cosmology, 2023.

Software I developed

4. pyIGIMF: for the efficient computation of the initial mass function in the context of the IGIMF theory (public code)
Please refer to the Accepted Publication No. 19 below

3. GalCEM: A detailed Galactic Chemical Evolution (public code)
Please refer to the Accepted Publication No. 7 below

2. Dust in Gadget3: Dust Evolution within a complex zoom-in simulation (private proprietary code)
Please refer to the Accepted Publication No. 3 below

1. Add-on: Improved Photometric Type-Ia Supernova Estimator for the Dark Energy Survey and the Legacy Survey of Space and Time (LSST) (private proprietary code)
Please refer to the Accepted Publication No. 1 below

Accepted Publications

20. The Initial Mass Function as the Equilibrium State of a Variational Process Gjergo, Zhang, Kroupa (2025)
Accepted for publication by Research in Astronomy and Astrophysics

19. Massive Star Formation at Supersolar Metallicities: Constraints on the Initial Mass Function Gjergo et al. (2025)
Accepted for publication by Research in Astronomy and Astrophysics

18. The effect of the environment-dependent stellar initial mass function on the baryonic Tully–Fisher relation Zonoozi et al. (2025)
Monthly Notices of the Royal Astronomical Society, Volume 542, Issue 3, pp. 1864-1872, 9 pp.

17. The impact of early massive galaxy formation on the cosmic microwave background Gjergo & Kroupa (2025)

16. Alien Type Ia Supernovae from the Milky Way Merger History and One Possible Candidate: Kepler's Supernova He, Zhou Gjergo, Fu (2025)

The Astrophysical Journal, Volume 986, Issue 2, id.123, 12 pp.

Gjergo is a corresponding author

15. The p-Laplacian as a framework for generalizing Newtonian gravity and Milgromian gravitation Scherer, Pflamm-Altenburg, Kroupa, Gjergo (2025)

Astronomy & Astrophysics, Volume 698, id.A167, 12 pp.

14. The MAGPI Survey – The kinematic morphology–density relation (or lack thereof) and the Hubble sequence at $z \approx 0.3$ Foster et al. (2025)

Publications of the Astronomical Society of Australia, Volume 42, id.e058, 22 pp.

13. The effect of the environment-dependent stellar initial mass function on the photometric properties of star-forming galaxies

Astronomy & Astrophysics, Volume 689, id.A221, 18 pp. Haslbauer et al. (2024)

12. First Detection of CO Isotopologues in a High-redshift Main-sequence Galaxy: Evidence of a Top-heavy Stellar Initial Mass Function

The Astrophysical Journal, Volume 970, Issue 2, id.136, 11 pp. Guo et al. (2024)

11. The Variation in the Galaxy-wide Initial Mass Function for Low-mass Stars: Modeling and Observational Insights Yan et al. (2024)

The Astrophysical Journal, Volume 969, Issue 2, id.95, 11 pp.

10. α -enhanced Astrochemistry: The Carbon cycle in extreme galactic conditions

Bisbas, Zhang, Gjergo, et al. (2024)

Monthly Notices of the Royal Astronomical Society, Volume 527, Issue 3, pp.8886-8906

9. Correction to: An analytical description of the evolution of binary orbital-parameter distributions in N-body computations of star clusters

Kroupa, Wang, Gjergo (2023)

Monthly Notices of the Royal Astronomical Society, Volume 526, Issue 4, pp.5777-5777

8. Assessing stellar yields in Galaxy chemical evolution: Observational stellar abundance patterns

Liang, Gjergo, Fan (2023)

Monthly Notices of the Royal Astronomical Society, Volume 522, Issue 1, pp.863-884

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7. GalCEM I: An Open-source Detailed Isotopic Chemical Evolution Code

Gjergo et al. (2023)

The Astrophysical Journal Supplement Series, Volume 264, Issue 2, id.44, 22 pp.

- 6. LAMOST meets Gaia: The Galactic open clusters** Fu et al. (2022)
Astronomy & Astrophysics, Volume 668, id.A4, 16 pp.
- 5. The MAGPI Survey – science goals, design, observing strategy, early results and theoretical framework** Foster et al. (2021)
Publications of the Astronomical Society of Australia, 2021, Volume 38, article id. e031
- 4. On the origin of dust in galaxy clusters at low-to-intermediate redshift** Gjergo et al. (2020)
Monthly Notices of the Royal Astronomical Society, Volume 493, Issue 2, p.2782-2792.
- 3. Dust evolution in galaxy cluster simulations** Gjergo et al. (2018)
Monthly Notices of the Royal Astronomical Society, Volume 479, Issue 2, p.2588-2606.
- 2. Analytic photometric redshift estimator for Type Ia supernovae from LSST** Wang, Gjergo, Kuhlmann (2015)
Monthly Notices of the Royal Astronomical Society, Volume 451, Issue 2, p.1955-1963.
- 1. Type Ia supernovae selection and forecast of cosmology constraints for the Dark Energy Survey** Gjergo et al. (2013)
Astroparticle Physics, Volume 42, p. 52-61.

Invited Talks

Seminar at Yunnan University, (Kunming, China)	Jan '25
Seminar at IBS, (Daejeon, South Korea)	Dec '24
Miniworkshop on GalCEM and GCE, (Beijing, China)	July '24
Seminar at Beihang University, (Beijing, China)	May '24
Origin of Elements and Cosmic Evolution (OECE2023, Beijing, CN)	Nov '23
Seminar at Tsinghua University, (Beijing, China)	May '23
Seminar at Bonn University, (Bonn, Germany)	Apr '22
Seminar at Charles University, (Prague, Czech Republic)	Mar '22
GW NEXT 2022, Online Conference (Beijing, CN)	Jan '22
National Astronomical Observatory of China Seminar (Beijing, CN)	May '21
Kavli Institute for Astronomy & Astrophysics Seminar (Beijing, CN)	Nov '20
Beijing Normal University Seminar (Beijing, China)	Nov '19
3 rd International Workshop on Physics (Tirana, Albania)	Oct '18
2 nd International Workshop on Physics (Tirana, Albania)	Sep '16
CARA outreach (Yerkes Observatory, WI, USA)	Jul '13

Technical Talks

MAGPI Busyweek (online, Australia)	June '21
Hydrodynamical Simulations Meeting (Trieste, Italy)	Sep '16
LSST-DESC Collaboration Meeting (Pittsburgh, PA)	Dec '13
LSST DESC Supernovae Working Group (11 Webinars)	Spring '17-'18

	DESSN Workshop (KICP at the University of Chicago, IL)	Jul '13
Talks and posters	Galaxies at Crossroads (Brno, Czech Republic)	Sep '24
	CosmoChem2024 (Palinuro, Italy)	Jun '24
	Nuclei in the Cosmos (NIC XVII, Daejeon, South Korea)	Sep '23
	Origin of Matter and Evolution of Galaxies OMEG16 (Hanoi, Vietnam)	Oct '22
	Interstellar Physics & Chemistry (Online conference, Zhuhai, China)	Jan '22
	Origin of Elements and Cosmic Evolution (Beijing, China)	Nov '19
	The Milky Way: LAMOST and other leading surveys (Yichang, China)	Oct '19
	Second Italian Cluster Conference (Naples, Italy)	May '18
	First Italian Cluster Conference (Turin, Italy)	Feb '17
	KROME Computational school (Florence, Italy)	Jun '16
	ICTP Workshop on Large-Scale Structure (Trieste, Italy)	Jun '16
	ICTP Summer School of Cosmology (Trieste, Italy)	Jun '16
	Lucchin astrophysics school (Naples, Italy)	May '16
	AAS 223 rd Meeting (Washington, D.C.) Chambliss Hon. Mention	Jan '14
	Santa Fe Cosmology Workshop (Santa Fe, NM)	Jul '13
	AAS 222 nd Meeting (Indianapolis, IN) Chambliss winner	Jun '13
	AAS 221 st Meeting (Long Beach, CA)	Jan '13
	ICTP Summer School of Cosmology (Trieste, Italy)	Jun '12
	Chicagoland and Midwest 1-Day Dark Matter Workshop (Chicago, IL)	Apr '12
	APS April Meeting (Atlanta, GA)	Apr '12